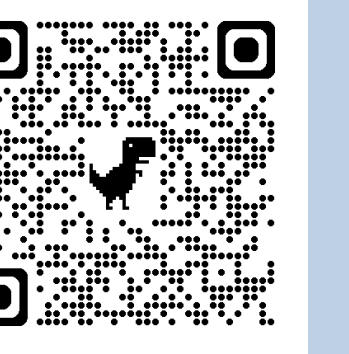


mztabm2cytoscape: Pathway Mapping and Visualization of Metabolomics Data

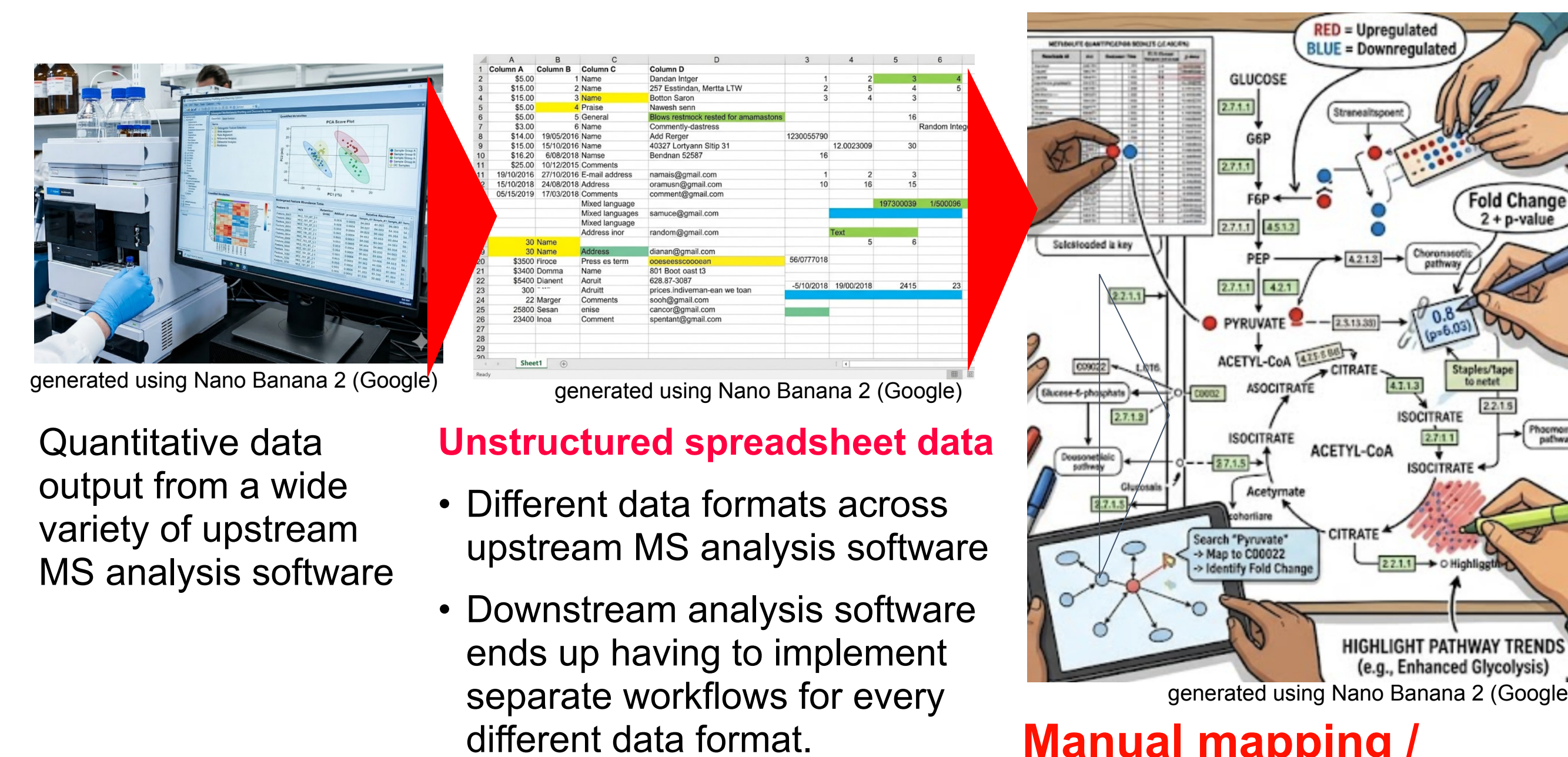
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mztabm2cytoscape automates the pathway mapping and visualisation of MS quantitative values, which is a representative downstream analysis in MS-based metabolomics.

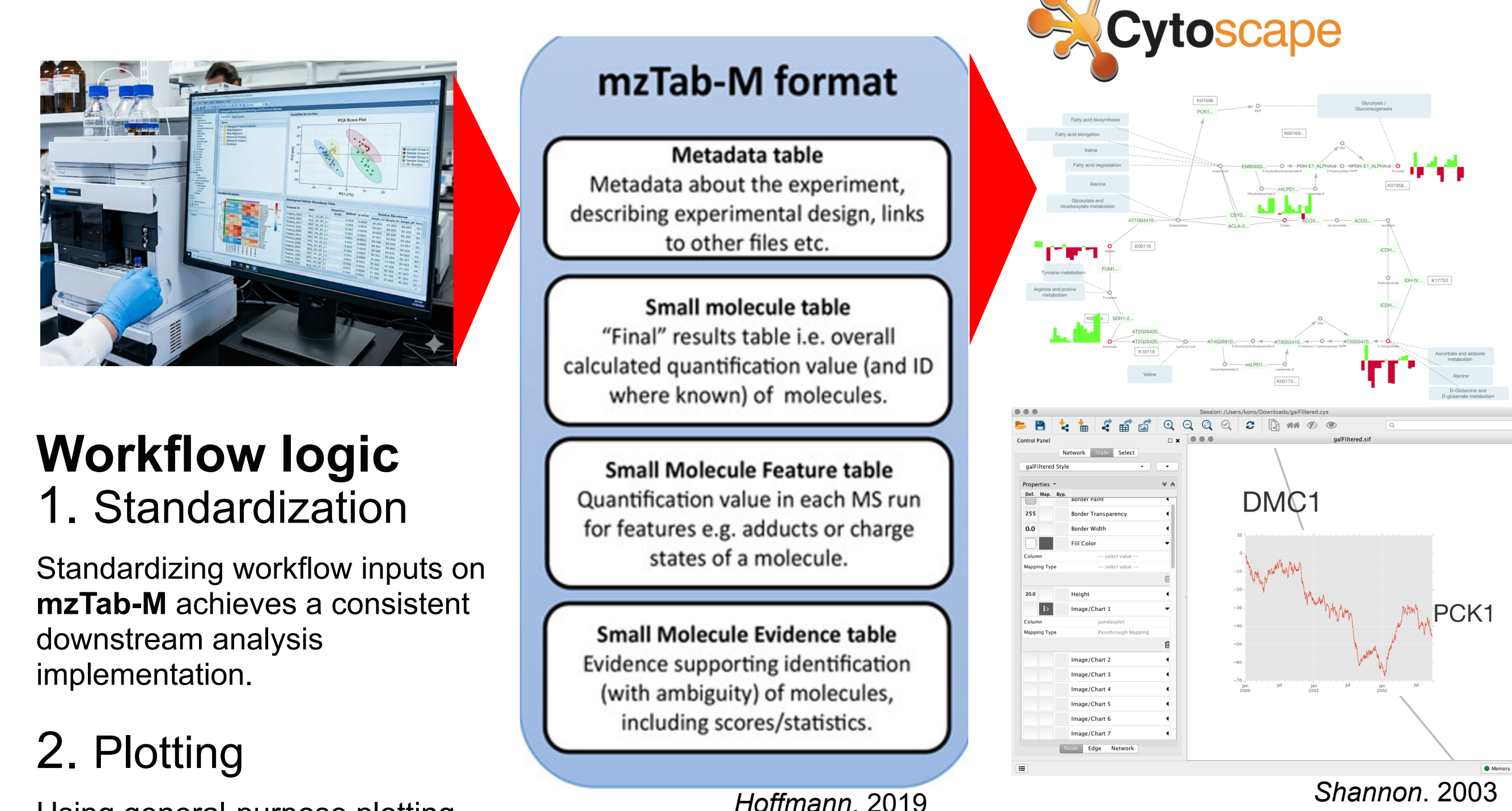
Background & Problem



The challenge

- Consolidating and simplifying the workflow
- Automating mapping and related steps
- a simple yet modular workflow design that enables diverse pathway visualizations

What we developed



mzTab-M

A data standard for sharing quantitative results in mass spectrometry metabolomics

The goal

Build a single workflow that can

- visualize outputs from any upstream metabolomics analysis software on pathways
- flexibly handle complex quantitative plots
- automatically harmonize small molecule IDs and integrate the plots of small-molecule quantitative values

Standardization

Standardizing workflow inputs on **mzTab-M** achieves a consistent downstream analysis implementation.

Plotting

Using general-purpose plotting libraries (e.g., **matplotlib**) allows for flexible and complex visualization of MS quantitative data.

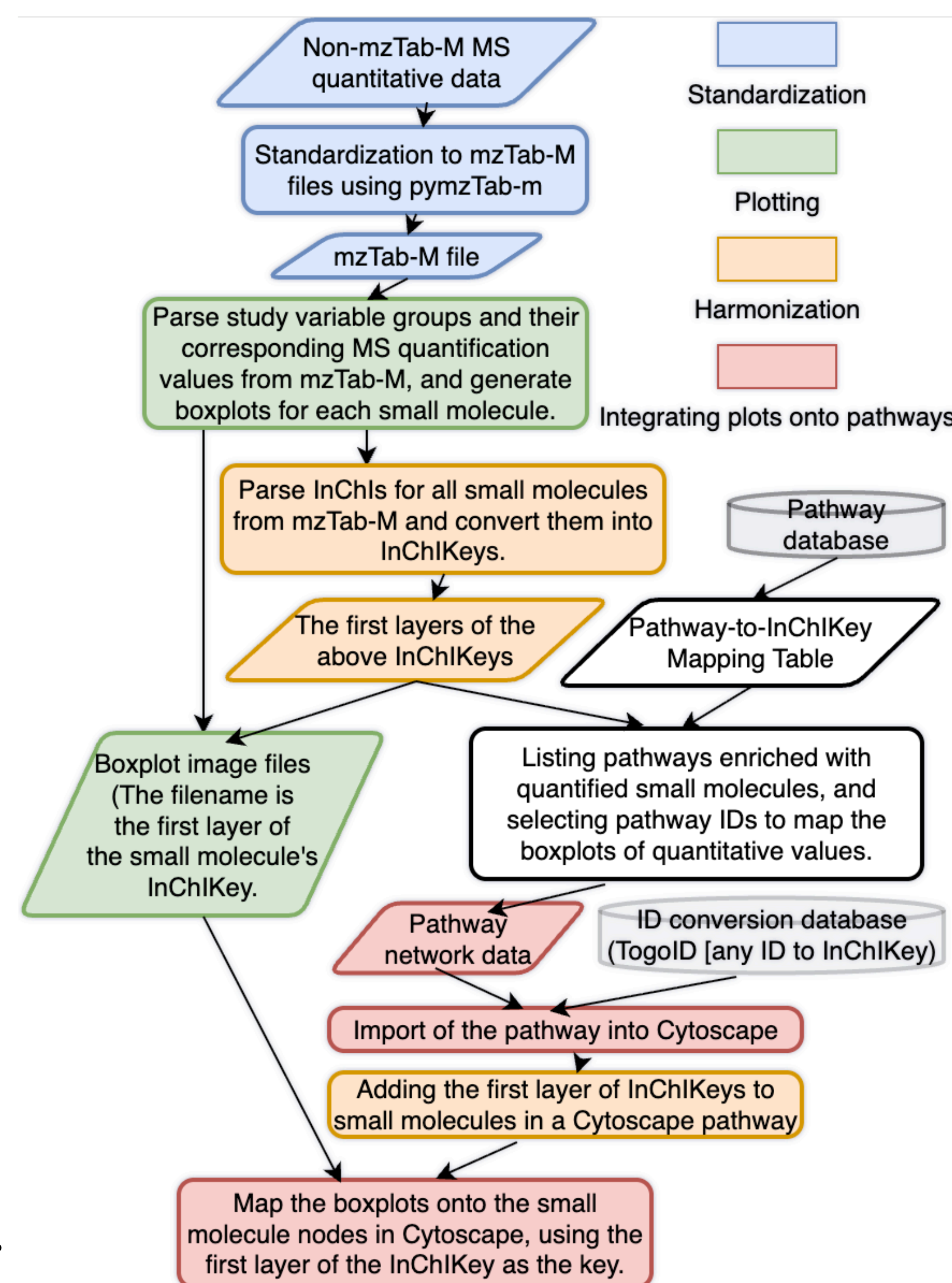
Harmonization

The **first block** of the **InChIKey** is utilized to harmonize small molecule IDs from MS quantification with those in pathways.

Integrating plots onto pathways

Cytoscape imports diverse pathway network data and can display plot images directly on the nodes.

Flowchart of mztabm2cytoscape



Visualization of metabolite abundances for Arabidopsis wild type and the two mutants

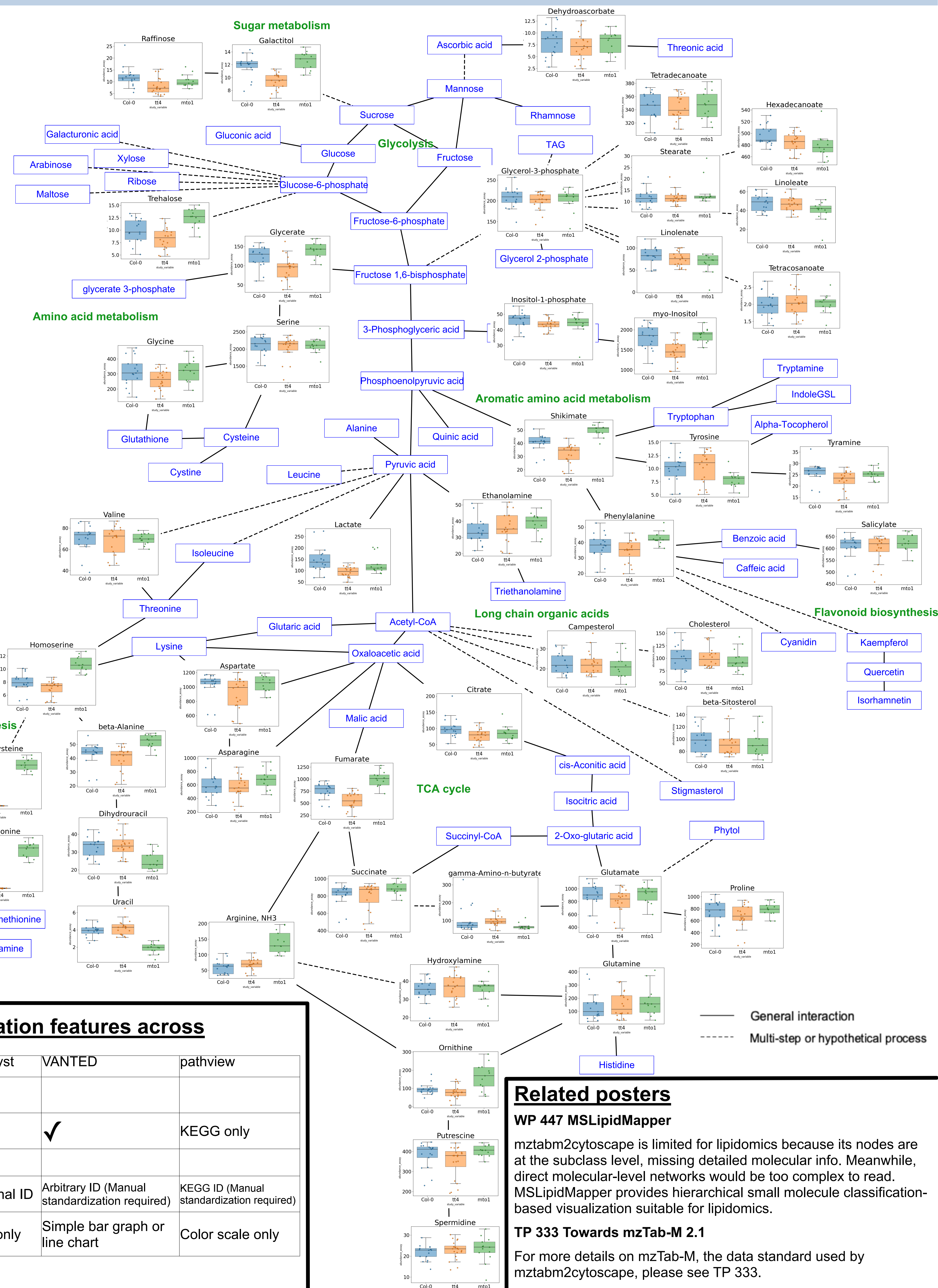
The pathway visualization on the right was automatically generated from the data below via the mztabm2cytoscape workflow.
(It is clearly visualized that the mto1 mutant overaccumulates Methionine.)

GC-MS quantitative data: **MTBLS40** ISA-TAB datasets from MetaboLights database
Pathway diagram data: **WP3622** GPML file from WikiPathways database (selected from the mapping enrichment results below [showing top 3]).

pathway	pathwayTitle	total_inchikeys	matched	ratio
WP3622	AtMetExpress overview	143	47	0.3286713
WP3604	Biochemical pathways: part I	419	34	0.0811455
WP3925	Amino acid metabolism	96	21	0.21875

Comparison of pathway mapping and visualization features across

	mztabm2cytoscape	MetaboAnalyst	VANTED	pathview
Use of mzTab-M for data integration with pathways	✓			
Data integration with any pathway data source	✓	KEGG only	✓	KEGG only
Automation of the workflow	✓	✓		
How small molecule IDs are standardized / mapped	First block of InChIKey	Its own internal ID	Arbitrary ID (Manual standardization required)	KEGG ID (Manual standardization required)
Visuals that can be mapped on the pathway nodes	Complex plots, such as boxplots, or anything that can be generated with matplotlib.	Color scale only	Simple bar graph or line chart	Color scale only



Related posters

WP 447 MS LipidMapper

mztabm2cytoscape is limited for lipidomics because its nodes are at the subclass level, missing detailed molecular info. Meanwhile, direct molecular-level networks would be too complex to read. MS LipidMapper provides hierarchical small molecule classification-based visualization suitable for lipidomics.

TP 333 Towards mzTab-M 2.1

For more details on mzTab-M, the data standard used by mztabm2cytoscape, please see TP 333.